



A3: 384-Node Supercomputing System

Huawei CloudMatrix384 SuperNode

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Huawei Tau (τ) Scaling Law



Introduction





CloudMatrix384
A Supernode for AI

遥遥领先
遥遥领先，未来已来！

- 1** 384 AI processors work together
- 2** Ultra-fast links help them talk quickly
- 3** Acts like one giant computer team
- 4** Built for training and running big AI models

Think of it like a huge team project:

many smart workers + one fast network + one big goal

Big, fast, and designed for the AI era.

> Key Idea: AI speed is not only about a powerful chip. It is also about how well many chips cooperate.

For Example:

- One student = slow for a giant task
- 384 AI workers + fast chatting = one super team
- CloudMatrix384 is the machine version of this teamwork



Meet the Team Inside CloudMatrix384

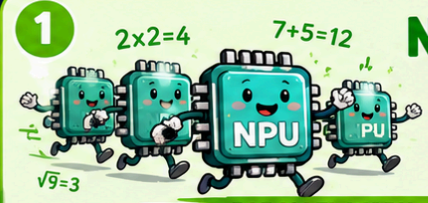


Think of the machine like a classroom project with workers, coaches, messengers, and notebooks.

1 NPU / AI Chips

The fast workers that do the heavy AI math.

PARALLEL WORK = SPEED!



2 CPU

The coach that organizes tasks and tells everyone what to do next.



3 NIC / Network Card

The super-fast messenger that helps machines talk to each other.



4 Memory

The shared notebooks where data and instructions can be stored and used quickly.



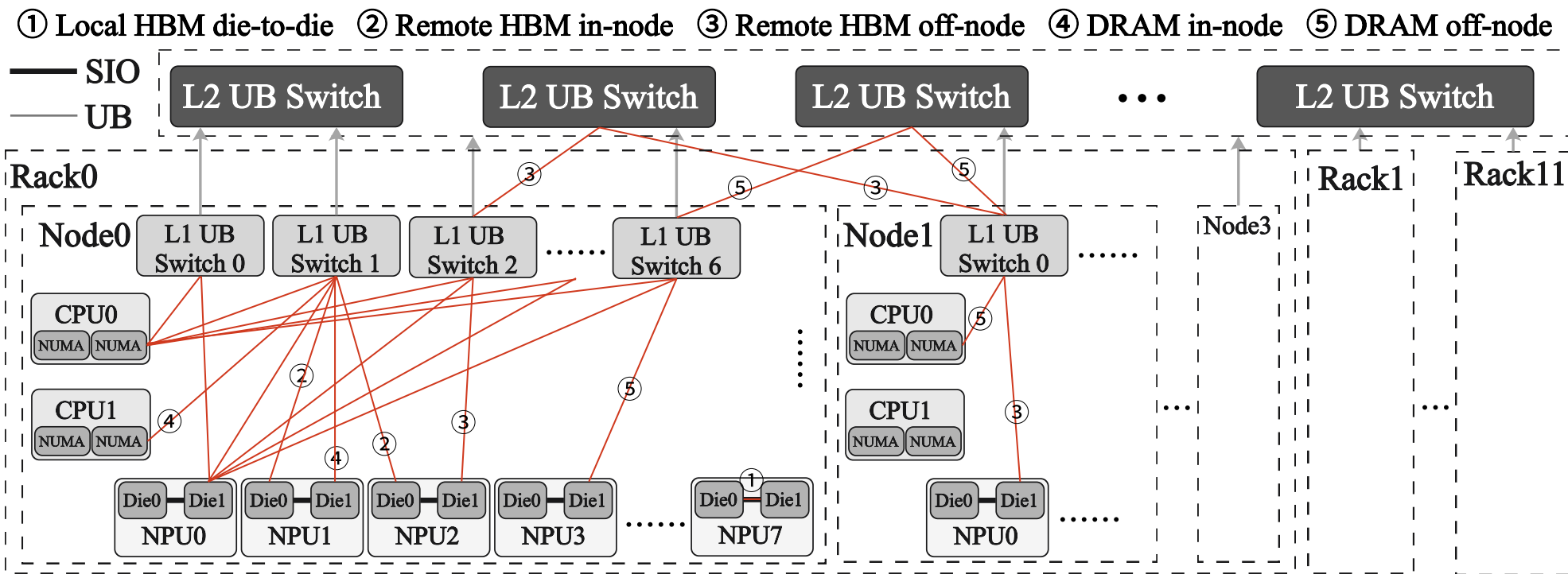
TOGETHER, WE GET THINGS DONE!

WORKERS + **COACH** + **MESSANGER** + **NOTEBOOKS** = **ONE GIANT AI TEAM!**



Beginner translation: many smart workers + a good coach + fast messengers + shared notes = one giant AI team.

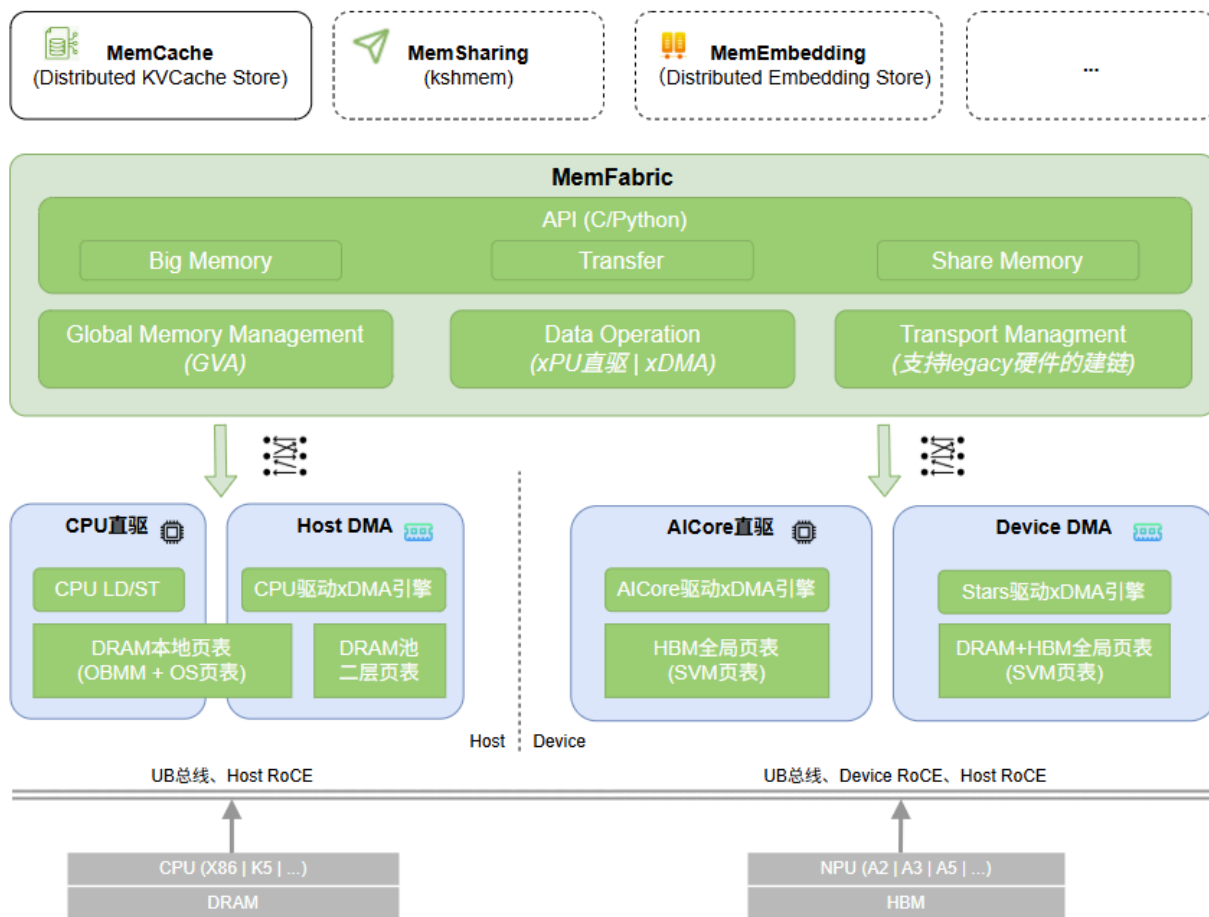




Design



Overview



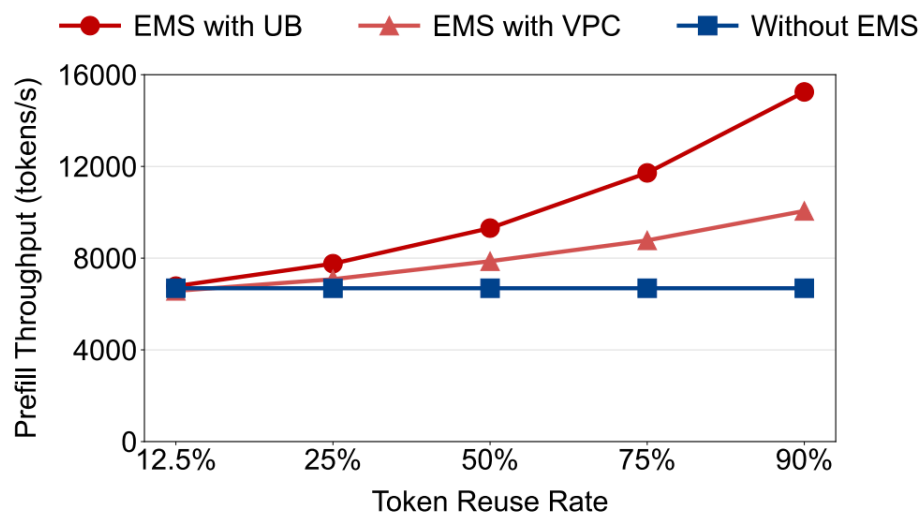


Experiment

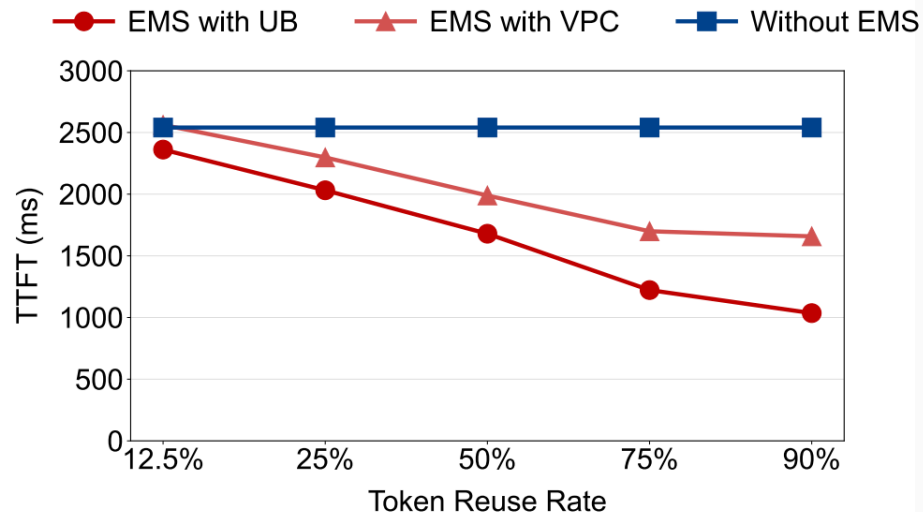


Table 2. Overall prefill throughput (per accelerator) for DeepSeek-R1.

Method	Batch Size	Input Length	Throughput (tokens/s)	Throughput per TFLOPS
DeepSeek on H800 (Blog)	N/A	N/A	4,026	2.03
SGLang on H100 (Default)	16,384	4,096	6,288	3.18
CloudMatrix-Infer (Default)	16,384	4,096	5,655	3.76
DeepSeek on H800 (Profile)	16,384	4,096	7,839	3.96
SGLang on H100 (Perfect EPLB)	16,384	4,096	7,417	3.75
CloudMatrix-Infer (Perfect EPLB)	16,384	4,096	6,688	4.45

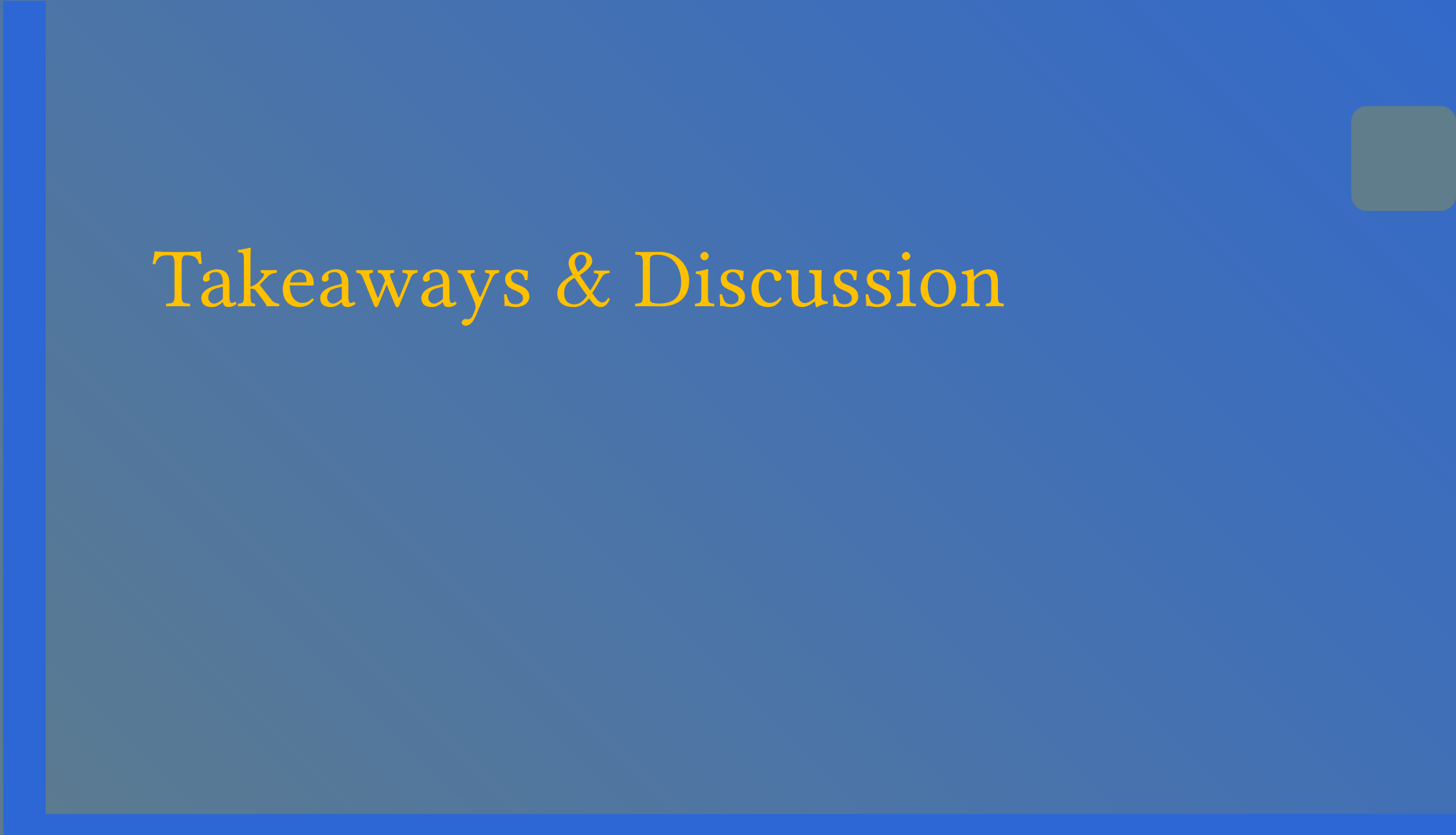


(a) Prefill throughput.



(b) TTFT.

Fig. 23. The overall prefill throughput and TTFT using EMS-Context Caching with different configurations.



Takeaways & Discussion

- Collectively, our findings validate Huawei CloudMatrix as a highly effective, scalable, and performance-optimized platform for deploying large-scale AI workloads, setting a benchmark for future AI datacenter infrastructures.

Takeaways & Discussion

- Why should non-CS students care?
 - New Competition
 - Countries and companies compete not only on chips, but also on system design and networking
 - Real Tradeoffs
 - More power also means more energy, cooling, cost, and engineering complexity

- For CS students,
 - Hardware Trends:
 - CloudMatrix384 focuses on full interconnectivity and resource pooling, not just adding more NPUs.
 - Abstraction Trends:
 - MemFabric uses GVA + xcopy to transform remote HBM/DRAM into memory resources that can be directly expressed by applications.
 - System Opportunities:
 - Memory-intensive workloads such as KV Cache, Embedding, and parameter caching will benefit first.
 - Research Space:
 - Placement, prefetching, consistency, and runtime scheduling still require workload-aware design.

Thank you!

https://gitcode.com/Ascend/memfabric_hybrid

Q & A